Midterm Study Guide

CSC 4780 – Network Security

Comprehensive Exam Review

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1. Networking Fundamentals

A **network** is a system of connected devices that exchange data.

A network can be modeled as a **graph** consisting of:

- **Nodes:** devices such as computers, phones, routers
- **Links:** connections between devices (wired or wireless)

Examples of links:

- Ethernet cables
- WiFi
- Bluetooth
- Radio signals

Many connected networks form the **Internet**.

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2. Network Architecture and Layering

Networking systems are complex, so they are divided into **layers**.

Benefits of Layering

- Simplifies system design
- Allows modular protocol development
- Hides implementation details
- Allows independent upgrades

Each layer provides **services** to the layer above it.

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3. TCP/IP Model

The Internet uses the **TCP/IP protocol stack**.

Layer	Example Protocols
Application	HTTP, SMTP, DNS
Transport	TCP, UDP
Network	IP
Link	Ethernet, WiFi

Data Units

Layer	Data Unit
Application	Data
Transport	Segment
Network	Packet
Link	Frame
Physical	Bits

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4. Ethernet and Link Layer

The **link layer** manages communication between directly connected devices.

Most common LAN technology:

Ethernet

Ethernet Frame Structure

- Synchronization bits
- Source MAC address
- Destination MAC address
- Payload (IP packet)
- Error detection

Ethernet allows devices to transmit at full speed but must handle **collisions**.

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5. Wireless Networking Problems

Wireless communication introduces challenges:

Hidden Terminal Problem

Example:

```
A ----> C <---- B
```

A and B cannot hear each other but both communicate with C.

Result: **collision at C**

Signal Fading

Wireless signals weaken with distance.

Interference

Multiple devices transmitting simultaneously cause corruption.

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6. Switching

A **switch** connects multiple LAN segments.

Switch Functions

- Forward frames
- Separate collision domains
- Filter network traffic

Forwarding Table Example

Destination	Port
A	3
B	0
C	3
D	3

Switches improve network efficiency.

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7. Internet Protocol (IP)

IP provides host-to-host communication across networks.

Example IP address:

```
129.82.138.254
```

Binary:

```
10000001.01010010.10001010.11111110
```

Important fact:

> An IP address identifies a **network interface**, not necessarily a device.

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8. Network vs Host Portion

Example:

```
1.1.2.1
```

Network portion:

```
1.1
```

Host portion:

```
2.1
```

This hierarchical structure enables **scalable routing**.

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9. IP Packet Structure

Important IP header fields:

- Version
- Header length
- Total length
- Identification
- Fragment offset

- TTL (Time To Live)
- Protocol
- Source address
- Destination address

TTL

TTL prevents packets from circulating forever in routing loops.

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10. Routing Concepts

Two key routing processes:

Forwarding

Selecting an **output port** based on the routing table.

Routing

Building the routing table that determines network paths.

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11. Autonomous Systems (AS)

An **Autonomous System (AS)** is a group of routers under a single administrative domain.

Routing types:

Routing Type	Protocol
Within AS	IGP
Between AS	BGP

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12. Border Gateway Protocol (BGP)

BGP connects autonomous systems on the Internet.

BGP is a **path vector protocol**.

Example path:

```
AS223 → AS225 → AS245 → AS7770
```

Loop Prevention

If a router sees its own AS number in the path, the route is rejected.

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13. BGP Security Issues

BGP originally had **no security mechanisms**.

Routers trust route advertisements.

This leads to attacks.

Common BGP Attacks

Prefix Hijacking

Attacker advertises IP space belonging to another network.

Subprefix Hijacking

Attacker advertises a **more specific prefix**.

More specific routes are preferred → traffic redirected.

Route Leak

A network advertises routes beyond their intended scope.

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14. RPKI (Resource Public Key Infrastructure)

RPKI verifies which AS is allowed to originate an IP prefix.

Key object:

ROA (Route Origin Authorization)

Contains:

- Authorized AS
- Prefix
- Maximum prefix length

Validation results:

- VALID
- INVALID
- NOT FOUND

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15. BGPSEC

BGPSEC secures routing paths.

Each AS:

1. Receives route update
2. Signs it cryptographically
3. Forwards signed update

Benefits:

- Prevents path manipulation
- Improves routing trust

Limitations:

- High computational cost
- Limited deployment

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16. DNS (Domain Name System)

DNS maps domain names to IP addresses.

Example:

```
google.com → 216.58.217.46
```

Humans remember **names**, computers use **numbers**.

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17. DNS Hierarchy

DNS is hierarchical.

```
Root
 └── TLD (.com, .edu)
```

```
        └── Domain
            └── Subdomain
```

Example:

```
www.cs.tntech.edu
```

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18. DNS Security Problems

Traditional DNS lacks authentication.

Attackers can forge DNS responses.

DNS Cache Poisoning

Goal:

Insert malicious DNS entries into resolver cache.

Result:

Users redirected to malicious servers.

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19. Kaminsky Attack

Attacker repeatedly queries random subdomains:

```
1.google.com
2.google.com
3.google.com
```

Eventually wins a race condition and poisons the DNS cache.

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20. DNSSEC

DNSSEC provides **cryptographic authentication** for DNS.

Provides:

- authentication
- integrity

Does NOT provide:

- confidentiality
- DDoS protection

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21. Public Key Infrastructure (PKI)

PKI manages digital certificates.

Certificates bind identities to public keys.

Certificate components:

- Public key
- Owner identity
- Validity dates
- CA signature

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22. Certificate Authorities (CA)

A **Certificate Authority** verifies identities and signs certificates.

Example statement:

```
"I certify that this public key belongs to Alice"
```

Browsers trust a list of root CAs.

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23. Infrastructure Trust vs Message Privacy

Different systems solve different security problems.

Technology	Purpose
DNSSEC	verify destination
BGPSEC	verify routing path
PKI	verify identity
Encryption	protect message

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24. Encryption

Encryption protects **confidentiality**.

Modern cryptography relies on computational hardness.

Examples:

- RSA
- AES
- Diffie-Hellman

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25. One-Time Pad (OTP)

OTP is the only system with **perfect secrecy**.

Requirements:

- Key must be random
- Key length = message length
- Key used only once

Encryption:

```
C = M ⊕ K
```

Decryption:

```
M = C ⊕ K
```

Security property:

```
P(M=m) = P(M=m | C=c)
```

Ciphertext reveals no information about plaintext.

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26. OTP Weakness

If a key is reused:

```
C1 = M1 ⊕ K
C2 = M2 ⊕ K
```

Then:

```
C1 ⊕ C2 = M1 ⊕ M2
```

This leaks information.

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27. Hash Functions

Hash functions map arbitrary input to fixed-length output.

Example:

```
SHA256(message) → digest
```

Desired Properties

- One-way
- Collision resistant
- Target collision resistant
- Pseudo-random

Applications

- Password storage
- File integrity verification
- Digital signatures

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28. Symmetric Key Encryption

Same key used for encryption and decryption.

Advantages:

- very fast
- efficient

Disadvantage:

- key distribution problem

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29. Block vs Stream Ciphers

Block Cipher

Encrypts fixed-size blocks.

Examples:

- DES
- AES

Typical block sizes:

```
64 bits
128 bits
```

Stream Cipher

Encrypts data **bit-by-bit or byte-by-byte**.

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30. Shannon's Cryptography Principles

Two key principles:

Confusion

Hide relationship between key and ciphertext.

Diffusion

Spread plaintext influence across ciphertext.

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31. Feistel Cipher Structure

Many block ciphers use the **Feistel network**.

Process:

1. Split plaintext into two halves
2. Apply round function
3. XOR result with other half
4. Swap halves
5. Repeat for multiple rounds

Decryption works by reversing subkeys.

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32. Data Encryption Standard (DES)

DES was historically the most widely used cipher.

Properties:

- Block size: **64 bits**
- Key size: **56 bits**
- Rounds: **16**

DES Components

- Initial permutation
- Expansion permutation
- S-boxes
- P-box
- Final permutation

S-boxes provide **non-linear substitution**.

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33. Avalanche Effect

Good encryption algorithms have the **avalanche effect**.

Changing one input bit changes roughly **half of output bits**.

This prevents attackers from guessing keys.

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34. DES Weakness

DES became insecure because of its small key size.

Example:

In 1998, the **Electronic Frontier Foundation** built hardware that cracked DES in about **56 hours**.

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35. Cryptanalysis Attacks

Differential Cryptanalysis

Examines how input differences affect ciphertext.

Requires chosen plaintext pairs.

Linear Cryptanalysis

Uses linear approximations of cipher operations.

Requires large numbers of known plaintexts.

Final Key Takeaways

Modern Internet security uses **layered protections**:

1. Infrastructure trust
    - DNSSEC
    - BGPSEC
2. Identity verification
    - PKI
3. Data protection
    - encryption
    - hashing

Security depends on:

- randomness
- key management
- computational hardness.